

Neural Network & Convolutional Neural Network

The goal of this assignment was to implement, train, and evaluate both a neural network (NN) and a convolutional neural network (CNN) using the USPS handwritten digits dataset. Both models were tested using stratified 5-fold cross-validation to provide consistent comparisons and evaluate model generalization.

PRE-PROCESSING

The dataset was loaded from OpenML. Since the USPS labels ranged from 0–10, samples labeled as "10" were removed, as the models were only designed to classify digits 0 through 9 (i.e., 10 classes).

To help the models train efficiently, the input features were standardized using `StandardScaler` (mean = 0, std = 1). This helps prevent features with larger scales from dominating the gradient updates. For the CNN, the input features were reshaped from flat vectors to 16×16 grayscale images with a single channel to match the expected input shape for `Conv2d`.

NEURAL NETWORK MODEL

The feedforward neural network was implemented using PyTorch and consisted of a 256-neuron input layer (one per pixel), followed by two hidden layers with 128 and 64 neurons respectively, both using ReLU activation. The final output layer had 10 neurons—one for each digit class. This architecture was chosen for its simplicity and effectiveness in handling flattened image data.

The activation function chosen was ReLU, as it tends to train faster and helps avoid vanishing gradients. The optimizer was Adam with a learning rate of 0.001, which is a common

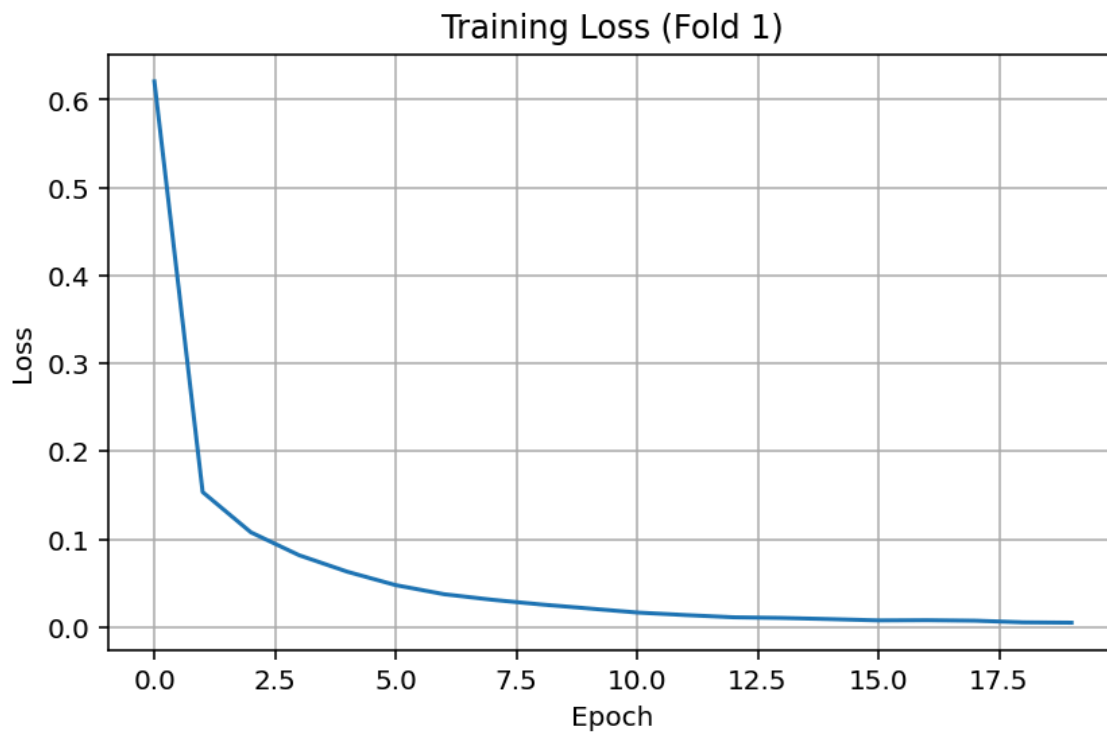
baseline for adaptive optimizers. The model was trained for 20 epochs with a batch size of 64. CrossEntropyLoss was used as the loss function, since it works well for multiclass classification problems.

This structure uses one Conv2D layer with 8 filters and a 3×3 kernel (padding=1) to extract local features, followed by 2×2 max pooling to downsample. Given the 16×16 image size, this balances spatial resolution and training efficiency. The final fully connected layers ($512 \rightarrow 64 \rightarrow 10$) allow for high-level abstraction without overfitting.

Each fold was trained independently and tracked for loss and accuracy. The neural network achieved an average accuracy of $97.33\% \pm 0.0024$ across five folds. The training loss curve showed smooth and steady convergence throughout all epochs, indicating stable learning behavior.

NN	Accuracy
Experiment 1	0.9770
Experiment 2	0.9741

Experiment 3	0.9740
Experiment 4	0.9717
Experiment 5	0.9699
Average	0.9733 ± 0.0024

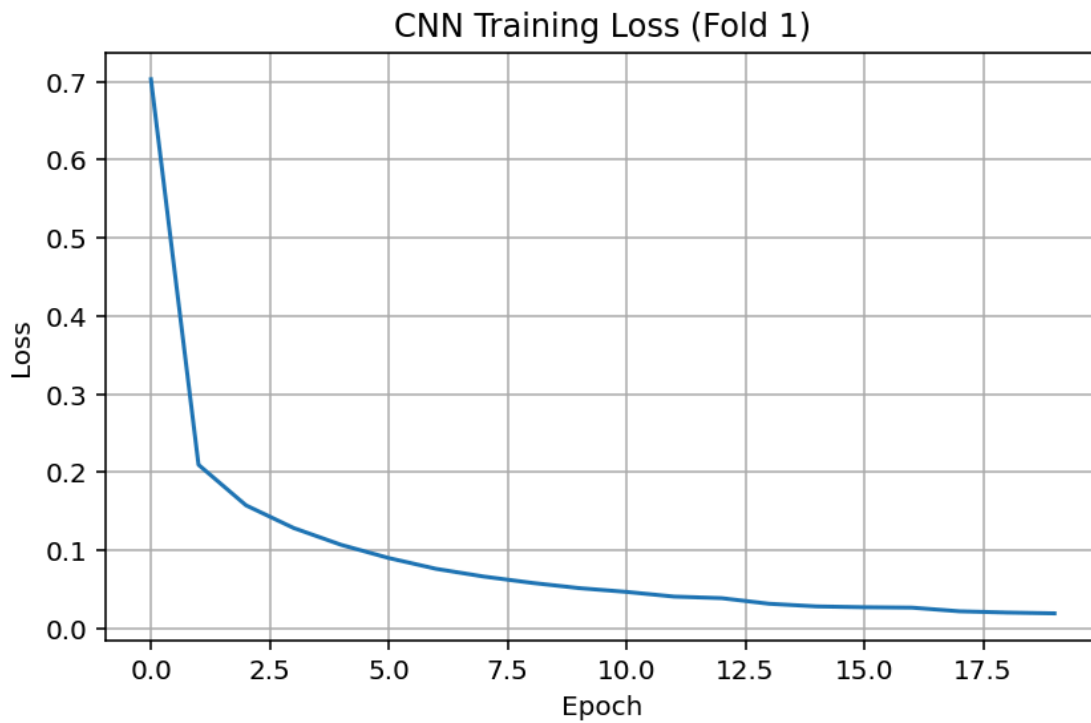


CONVOLUTIONAL NEURAL NETWORK MODEL

The CNN was kept minimal to suit the small 16×16 grayscale images. It included a single convolutional layer with 8 filters (3×3 kernel, padding=1), followed by ReLU activation and 2×2 max pooling. The output was then flattened and passed through a fully connected layer ($512 \rightarrow 64$) before the final classification layer with 10 outputs.

The architecture was chosen to retain enough spatial detail while being light enough to train quickly. It uses the same training parameters as the NN (20 epochs, learning rate = 0.001, batch size = 64). The model converged quickly and demonstrated strong performance.

The convolutional neural network reached a slightly higher average accuracy of 97.66% ± 0.0046 . Its training loss consistently decreased over the epochs, and the learning curve reflected strong convergence without instability or overfitting, particularly evident in the first fold.



PERFORMANCE: NN V. CNN

Both models achieved high accuracy on the USPS dataset, but the CNN slightly outperformed the NN across all folds. While the difference in average accuracy is relatively small, CNN's slight performance advantage is expected with image data, as the convolutional layer is better at capturing spatial patterns and local features (like edges or corners) than a densely connected NN.

NN	Accuracy	CNN	Accuracy
Experiment 1	0.9770	Experiment 1	0.9800
Experiment 2	0.9741	Experiment 2	0.9805
Experiment 3	0.9740	Experiment 3	0.9740
Experiment 4	0.9717	Experiment 4	0.9799
Experiment 5	0.9699	Experiment 5	0.9687
Average	0.9733 ± 0.0024	Average	0.9766 ± 0.0046

CONCLUSION

This assignment demonstrated that both NN and CNN models can effectively classify USPS digits. The NN achieved a strong 97.33% accuracy, while the CNN slightly outperformed it at 97.66%, likely due to its ability to extract spatial features. The results confirm CNNs are better suited for visual data. Training curves for both models showed smooth convergence, supporting the chosen architectures and hyperparameters.